Quiz 1

CS 102 Computer Programming I · Fall Semester 2026
Week 4 · Duration: **35 MINUTES** · Version **B**

New Uzbekistan University

FULL NAME: _____ ID: _____ GROUP: _____

Section	Points	Score
A Multiple choice	30	
B Fill in the gaps	40	
C Trace and write	30	
Total	**100**	

Rules. No notes, phones or computers. Write answers *only* in the space given. For fill-in-the-gap questions write exactly one piece of C++ code in each numbered gap — a keyword, an operator, a type or an expression. Assume every code fragment is inside `main` with `#include <iostream>` and `using namespace std;`.

Section A: Multiple Choice (30 points)

Circle **one** answer for each question.

A1. (6 points) The hexadecimal literal `0x3C` has the decimal value: A. 36 B. 60 C. 54 D. 72

A2. (6 points) What does `cout << 9 / 4 * 2.0 + 9 % 4;` print? A. 5.5 B. 6 C. 5 D. 4

A3. (6 points) What does this code print?

```
int x = 3;
if (x > 5 && x++ > 0) {
    cout << "yes ";
}
cout << x;
```

A. `yes 3` B. `yes 4` C. `3` D. `4`

A4. (6 points) Which expression does **not** evaluate to `3.5`?

A. `7 / 2.0`

B. `static_cast<double>(7) / 2`

C. `static_cast<double>(7 / 2)`

D. `7.0 / 2`

A5. (6 points) What does this print? `char c = 'a' + 3; cout << c << (c + 1);`
Hint: in the ASCII table, `'a'` *= 97.* A. `de` B. `d101` C. `100101` D. `d 101`

SIGNATURE: _____ ID: _____

Section B: Fill in the Gaps (40 points)

Each numbered gap is worth 2 points. Every program must compile and produce the output shown. Write your answers in the boxes under each program.

B1. (10 points) **Minutes to days, hours, minutes.** Input `3725` must print `2 d 14 h 5 min`.

```
#include <iostream>
using namespace std;
int main() {
    (1)_____ total;
    cin >> total;
    int d = total (2)_____ 1440;
    int h = (total (3)_____ 1440) / 60;
    int m = total (4)_____ 60;
    cout << d << " d " << h << " h " << m << " min" << (5)_____;
    return 0;
}
```

(1) _____ (2) _____ (3) _____ (4) _____ (5) _____

B2. (10 points) **Average of two tests.** `PASS` if the average is at least 60, otherwise `FAIL`; an average outside 0–100 is invalid. Input `70 55` must print `Average: 62.5, rounded: 63, PASS`.

```
int t1, t2;
cin >> t1 >> t2;
int sum = t1 + t2;
double avg = (6)_____(sum) / 2;    // must NOT be integer division
(7)_____ passed = avg (8)_____ 60;
if ((9)_____) {             // outside 0..100
    cout << "Invalid" << endl;
} else {
    cout << "Average: " << avg << ", rounded: " << (10)_____(avg + 0.5);
    if (passed) {
        cout << ", PASS" << endl;
    } else {
        cout << ", FAIL" << endl;
    }
}
```

(6) _____ (7) _____ (8) _____ (9) _____ (10) _____

B3. (10 points) **FizzBuzz for one number.** Divisible by 3 and 5 prints `FizzBuzz`; only by 3, `Fizz`; only by 5, `Buzz`; otherwise the number. Input `15` prints `FizzBuzz`, `9` prints `Fizz`, `7` prints `7`.

```
int n;
cin >> n;
bool by3 = n (11)_____ 3 == 0;
bool by5 = n % 5 (12)_____ 0;
if (by3 (13)_____ by5) {
    cout << "FizzBuzz" << endl;
} (14)_____ (by3) {
    cout << "Fizz" << endl;
} else if ((15)_____) {
    cout << "Buzz" << endl;
} else {
    cout << n << endl;
}
```

(11) _____ (12) _____ (13) _____ (14) _____ (15) _____

B4. (10 points) **Disk usage (watch the total!).** A folder has up to 10^5 files of up to 10^5 bytes each. It is *large* if the total is at least 10^9 bytes and has *many* files if there are at least 1000. Input `30000 100000` must print `Both`.

```
int count, size;
cin >> count >> size;
(16) _____ total = static_cast<long long>(count) * size;
bool large = total >= 1000000000;
bool many = count >= 1000;
if (large (17) _____ many) {
    cout << "Both" << endl;
} else if ((18) _____) {
    cout << "Large" << endl;
} else if (many) {
    cout << "Many" << endl;
} (19) _____ {
    cout << "Small" << endl;
}
```

(16) [ ] (17) [ ] (18) [ ] (19) [ ]

(20) Why would `int total = count * size;` print `Many` for the input above?

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Section C: Trace and Write (30 points)

C1. (15 points) Write **exactly** what this fragment prints (mind the spaces).
Hint: in the ASCII table, 'a' = 97 (and 'A' = 65).

```
int a = 9, b = 4, c = 0;
double d = a / b + a % b / 4.0;
if (a < b)
    if (c == 0 || a / c > 1)
        cout << "X";
else
    cout << "Y";
cout << "d=" << d;
char ch = 'Z' - b;
if (ch >= 'A' && ch <= 'Z')
    ch = ch + 32;
cout << " " << ch << " " << (ch < 'w') << endl;
```

C2. (15 points) **Total time of two days.** Write a program that reads the time a student studied on Monday and on Tuesday, each as hours and minutes (`h1 m1 h2 m2`), and prints the total as hours and minutes. Use only `int`, `/` and `%`.

```
3 50 2 25   → 6 h 15 min
0 45 0 20   → 1 h 5 min
1 0 2 30    → 3 h 30 min
```

END OF QUIZ